MAI5043: Applied Machine Learning

Machine Learning Regression Assignment

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1. Dataset Overview

This project predicts residential property sale prices using the Kaggle House Prices – Advanced Regression Techniques dataset [1]. The dataset describes houses sold in Ames, Iowa, USA and contains structural, location-related and condition-related attributes. SalePrice is a continuous numerical target, so the problem is a supervised regression task. The aim was to explore the data, prepare reliable input features, train an appropriate regression algorithm, evaluate predictions using unseen labelled examples, and improve the model through cross-validated tuning.

Table 1. Original dataset and modelling setup

Item	Verified value / description
Source	Kaggle: House Prices – Advanced Regression Techniques
train.csv	1,460 rows × 81 columns (includes Id and SalePrice)
test.csv	1,459 rows × 80 columns; SalePrice is absent
Available predictors	79 original predictors after removing Id and SalePrice
Original predictor types	36 numerical and 43 categorical
Model input / target	Property attributes → SalePrice (USD)
Held-out split	80% train (1,168); 20% held-out test (292); random_state = 42

1.1 Target variable and key attributes

The SalePrice variable has mean $180,921.20, median $163,000, standard deviation $79,442.50, and ranges from $34,900 to $755,000. The mean exceeding the median suggests a higher-price tail, examined further in Section 2. The target contains no missing observations. Important available predictors include OverallQual (overall material and finish quality), GrLivArea (above-ground living area in square feet), GarageCars (garage capacity), TotalBsmtSF (basement floor area), YearBuilt, and Neighborhood (an Ames location category).

1.2 Labelled validation versus Kaggle predictions

The original Kaggle test.csv has no true SalePrice values, so it cannot be used to calculate MAE, RMSE or R^2. Therefore, train.csv was split into a fitting portion and a labelled 292-house held-out evaluation portion. The Kaggle test.csv was used only at the end to generate 1,459 predictions for the submission file. All performance metrics below refer to the held-out labelled split unless explicitly identified as cross-validation (CV) results.

2. Data Analysis

2.1 Exploratory analysis approach

Exploratory data analysis (EDA) assessed the target distribution, relationships between price and predictors, correlations among candidate features, missing-value patterns, and potentially unusual properties. A histogram was chosen to identify target skewness; scatter plots were used for continuous predictors; a boxplot summarized price across ordered quality groups; and a Pearson heatmap provided a compact comparison of linear associations. These visualizations are relevant to regression because they help identify useful predictors, nonlinearity, skewness and observations that might generate large prediction errors.

2.2 Sale-price distribution

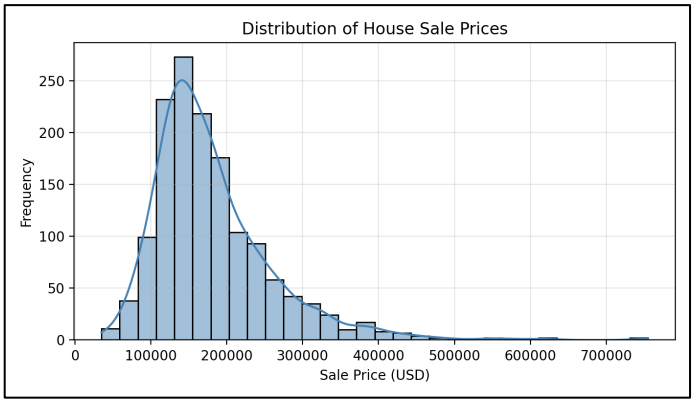


Figure 1. Distribution of observed sale prices.

Figure 1 reveals a right-skewed price distribution (skewness = 1.8829). Most houses fall in lower or middle price ranges, while a small number of properties exceed $400,000 and extend towards $755,000. As a result, squared-error measures such as RMSE can be disproportionately influenced by inaccurate predictions on expensive houses. The data also suggest that an average numerical error alone may not describe performance uniformly across the price range.

2.3 Material and finish quality

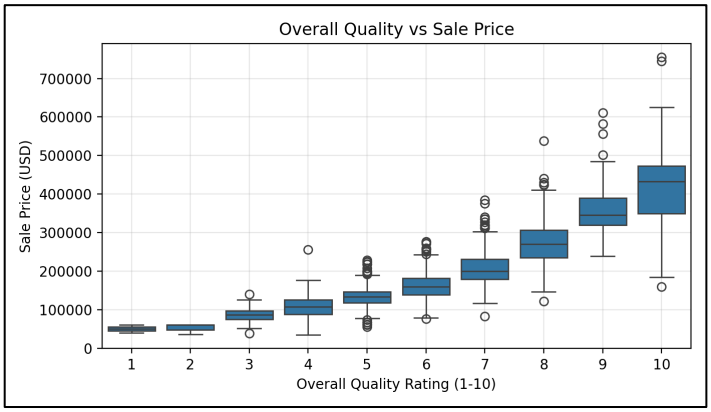


Figure 2. Price by overall quality rating.

Figure 2 shows a clear increase in median SalePrice with OverallQual, a rating from 1 (very poor) to 10 (very excellent). Median price rises from $50,150 at quality 1 and $133,000 at quality 5 to $269,750 at quality 8 and $432,390 at quality 10. There is still considerable within-category variation, particularly for high-quality properties, so a quality rating alone cannot determine sale price. Its strong relationship makes it a valuable candidate predictor in combination with other property attributes.

2.4 House size, correlation and unusual observations



Figure 3. Above-ground living area and sale price.

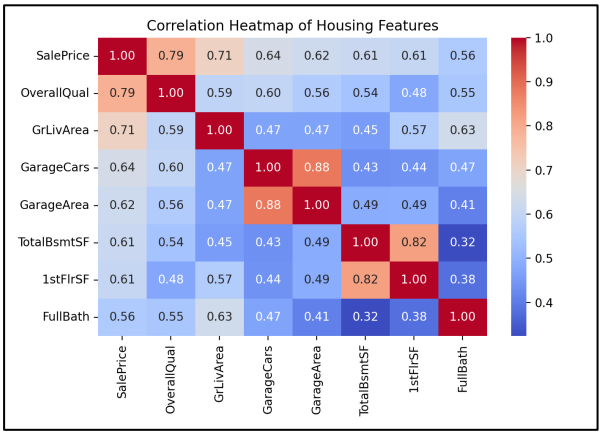


Figure 4. Pearson correlation heatmap.

Figure 3 illustrates a strong positive relationship between above-ground living area (GrLivArea) and SalePrice, with a Pearson correlation coefficient of 0.7086. The scatter plot is appropriate for regression analysis because it shows how house prices vary with a continuous predictor and helps identify unusual observations. Generally, larger properties achieve higher selling prices, although the variation in prices increases as living area expands. Four properties exceed 4,000 sq ft, including two unusually large houses of 4,676 and 5,642 sq ft sold for $184,750 and $160,000, respectively. These observations were retained rather than automatically removed because they may represent genuine transactions and provide useful information for model training.

Figure 4 presents the Pearson correlation coefficients between SalePrice and selected numerical housing features. OverallQual (r = 0.791) has the strongest positive correlation with SalePrice, followed by GrLivArea (r = 0.709), GarageCars (r = 0.640), and GarageArea (r = 0.623). The heatmap also identifies a strong correlation between GarageCars and GarageArea (r = 0.88), suggesting that these variables contain overlapping information. This visualization is useful for identifying potential predictors and relationships between features.

2.5 Quantitative correlation findings

Table 2. Selected correlations with SalePrice from the training dataset

Feature	Pearson r	Interpretation
OverallQual	0.791	Strong positive association with quality rating
GrLivArea	0.709	Larger above-ground living area generally sells higher
GarageCars	0.640	Greater garage capacity generally associates with price
GarageArea	0.623	Larger garage floor area generally associates with price
TotalBsmtSF	0.614	More basement floor area generally associates with price
FullBath	0.561	More full bathrooms generally associate with price

Table 2 summarizes the six selected numerical features positively correlated with SalePrice. The findings indicate that property quality, living space, and available facilities are relevant factors in house price prediction. These relationships support the selection of Random Forest Regression, which can model nonlinear relationships and interactions among multiple predictors.

3. Preprocessing Steps

3.1 Data quality checks and missing values

The notebook found zero duplicated rows, both with the unique Id and after excluding Id, so no records were removed for duplication. Missing values were unevenly distributed across the features. A critical distinction was made between genuinely unknown information and a feature that is absent because the house does not have that facility. The supplied data_description file identifies NA as “No Pool”, “No Garage”, “No Basement” or “No Alley” for several categorical features; these cases should not automatically be treated as data-entry errors.

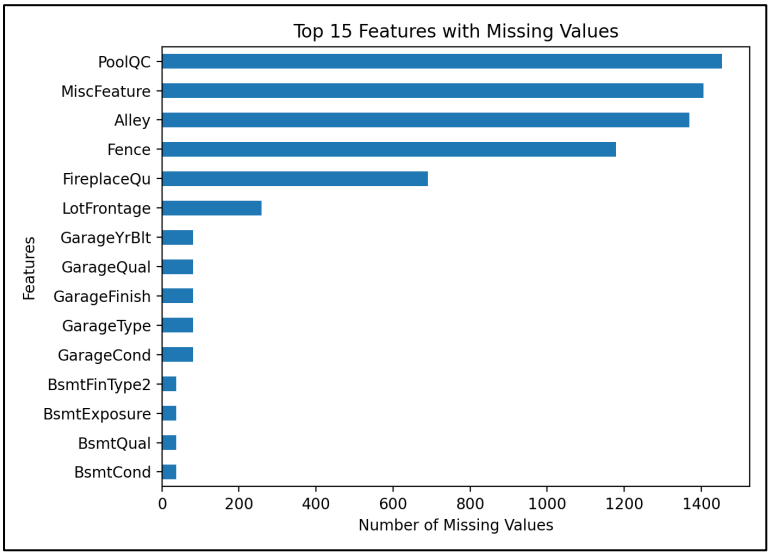


Figure 5. Fifteen original training features with the most missing values.

Table 3. Highest missing-value counts in the supplied training data

Feature	Missing	Percentage	Interpretation / response
PoolQC	1,453	99.52%	Often no pool; preserve absence as a category
MiscFeature	1,406	96.30%	Often no miscellaneous feature
Alley	1,369	93.77%	Often no alley access
Fence	1,179	80.75%	Often no fence
FireplaceQu	690	47.26%	Often no fireplace
LotFrontage	259	17.74%	Numerical; impute from training median

3.2 Pipeline design and missing-data treatment

A reproducible scikit-learn Pipeline applies feature engineering, then ColumnTransformer preprocessing, then the estimator [2]. Features that represent absent amenities (including Alley, selected basement, garage and fireplace variables, PoolQC, Fence and MiscFeature) are filled with the literal category “None”. For MasVnrType, the existing “None” category means no masonry veneer and is preserved; genuinely missing entries are instead labelled “Unknown”. All remaining numerical missing values are filled using the training-set median, with SimpleImputer(add_indicator=True) adding missingness indicators. Remaining categorical gaps are filled using the most frequent training category.

3.3 Encoding and feature engineering

Categorical variables were transformed with OneHotEncoder (handle_unknown="ignore", min_frequency=10). This avoids assigning an artificial numeric order to unordered categories, tolerates unseen levels in evaluation data, and groups infrequent levels. The MSSubClass numerical codes were converted to strings because they describe dwelling types rather than a continuous measurement. As implemented in the notebook, 79 original predictors became 85 after six feature-engineering additions and 276 processed model columns after imputation/encoding.

Table 4. Engineered features and purpose

Derived feature	Calculation / meaning	Potential value
HouseAgeAtSale	YrSold − YearBuilt	Age at transaction time
YearsSinceRemodel	YrSold − YearRemodAdd	Recency of renovation
TotalAboveGroundAndBasementArea	GrLivArea + TotalBsmtSF	Combined usable-size proxy
TotalBathrooms	FullBath + 0.5×HalfBath + BsmtFullBath + 0.5×BsmtHalfBath	Combined bathroom provision
HasGarage	1 when GarageType is not “None”, otherwise 0	Explicit garage availability
MissingLotFrontage	1 when LotFrontage is missing, otherwise 0	Flags unavailable frontage data

3.4 Scaling, outliers and leakage prevention

No normalization or standardization was used for the final Random Forest model because splits are determined by value order rather than absolute scale. The extreme living-area observations identified in EDA were kept; no unverified deletions were made. The 80:20 train/test split was created before fitting imputers and encoders, and these transformations were fitted on the training subset only. Cross-validation later re-fitted the complete pipeline within each fold, helping protect against leakage of validation information into preprocessing [3].

4. Model Design

4.1 Selection of Random Forest Regression

Random Forest Regression was chosen as the main model because house prices may depend on nonlinear relationships between property features. For example, the relationship between living area and price may vary depending on the property's quality, age, and neighborhood. Random Forest combines predictions from multiple decision trees trained on different bootstrap samples [4]. By averaging these predictions, the model can reduce the variation associated with a single decision tree while capturing relationships between features without specifying them in advance.

4.2 Original model configuration

Table 5. Original Random Forest implementation

Setting	Value	Reason
Algorithm	RandomForestRegressor	Nonlinear, interaction-sensitive regression
n_estimators	200	Ensemble of 200 trees for stable averaging
max_depth	None (default)	Trees may grow until other stopping conditions
min_samples_leaf	1 (default)	Original unrestricted leaf-size baseline
max_features	1.0 (default)	All features considered at each split
random_state	42	Reproducible bootstrap sampling and splits
n_jobs	−1	Use available cores during fitting

4.3 Linear Regression reference and workflow

Linear Regression was also trained as a benchmark using the same feature engineering and preprocessing pipeline. Unlike Random Forest, it estimates house prices using a weighted combination of the processed input features. Comparing both models on the same 292-house held-out test set helps determine whether Random Forest provides better predictions for this dataset.

Table 6. Implemented machine learning workflow

Stage	Application in the notebook
Input → EDA	Inspect train/test shapes, statistics, skewness, correlations and missingness
Split → preprocess	1,168 fitting records and 292 held-out records; train-fitted transformations
Train → predict	Linear Regression reference and 200-tree Random Forest baseline
Evaluate → optimize	MAE, MSE, RMSE, R^2; 3-fold CV with GridSearchCV
Finalize → export	Selected Random Forest; error analysis and Kaggle predictions

4.4 Strengths and limitations of model choice

Random Forest can handle nonlinear relationships and interactions between housing features without requiring feature scaling. However, its predictions are more difficult to interpret than those of Linear Regression, and it may struggle to predict prices beyond the range represented in the training data. Correlated features can also affect the interpretation of feature-importance scores. To examine these limitations, the

model was assessed using residual plots, training and testing performance, and prediction errors across different house-price groups.

5. Evaluation Results

5.1 Hold-out evaluation and metric definitions

The two reference models were fitted on the same 1,168-house training subset and evaluated on the same 292 held-out labelled houses. This design ensures that the comparison uses identical evaluation cases. Four metrics were used because they describe different aspects of regression performance rather than relying on one summary score.

Table 7. Regression metrics used and interpretation

Metric	Formula / meaning	Interpretation
MAE	Mean of |actual − predicted|	Average absolute error in USD; lower is better
MSE	Mean of (actual − predicted) ²	Squares large errors; measured in USD²
RMSE	Square root of MSE	Penalizes large deviations; measured in USD
R²	1 − SSE/SST	Relative explanatory performance; higher is better

5.2 Linear Regression versus original Random Forest

Table 8. Results on the same 292-house held-out test split

Model	MAE (USD)	MSE (USD²)	RMSE (USD)	R²
Linear Regression	20,692.57	999,288,376	31,611.52	0.8697
Original Random Forest	17,433.97	865,449,435	29,418.52	0.8872

The original Random Forest outperformed Linear Regression on every reported test metric. Its RMSE was $2,193.00 lower and its MAE was $3,258.60 lower than the linear reference. The Random Forest R^2 of 0.8872 indicates that, on this held-out split, its predictions account for about 88.7% of the variation in observed sale prices relative to a constant-mean baseline. The remaining errors are still material in dollar terms, particularly for high-value homes.

5.3 Training versus testing performance

Table 9. Original Random Forest generalization check

Metric	Training	Held-out test
MAE (USD)	6,214.50	17,433.97
R²	0.9801	0.8872

Training performance was substantially stronger than held-out performance. The train/test MAE gap was $11,219.47, while R^2 fell from 0.9801 to 0.8872. This difference suggests that the flexible original forest fits the observed training properties much more closely than unseen houses. However, a training–testing gap alone is not evidence that the model is unusable; it motivates the subsequent regularization-oriented hyperparameter search and a more detailed assessment of residuals.

5.4 Prediction agreement and residuals

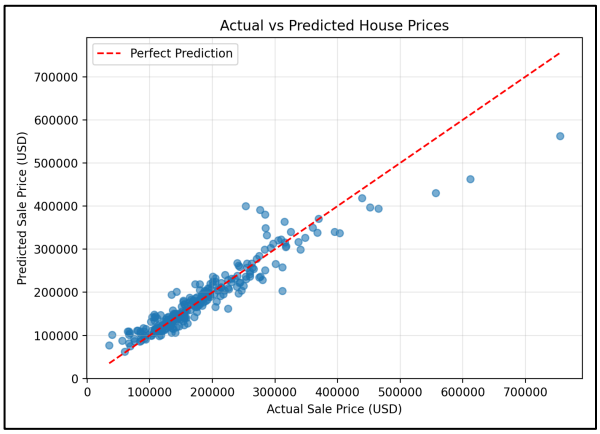


Figure 6. Original forest: actual versus predicted.

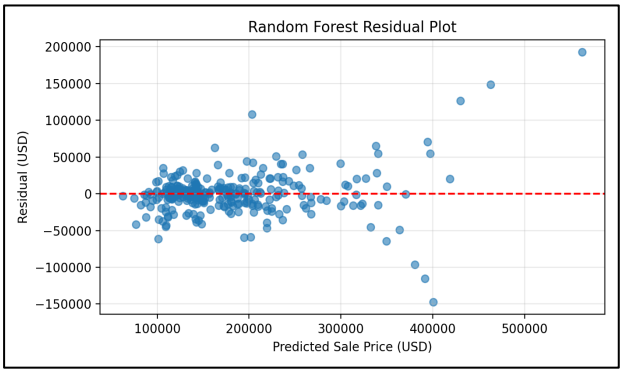


Figure 7. Original forest: residuals versus prediction.

In Figure 6, the red dashed diagonal represents perfect prediction. Lower- and middle-price observations lie relatively close to it, showing that the model captures the overall increasing pattern between true and predicted sale prices. Nevertheless, several high-value houses fall well below this diagonal, showing underprediction of some expensive properties. Predictions also appear less dispersed than actual high-end prices, a typical constraint of averaging tree predictions.

For Figure 7, residuals are defined as actual price minus predicted price. Positive residuals mean that the model underpredicted; negative residuals mean that it overpredicted. Residuals for many moderate-price houses cluster near zero, while the spread becomes much larger for some expensive predictions. The residual pattern is not uniform across the price range, so the global error metrics should be supplemented with price-group analysis rather than interpreted as the same error for every segment.

5.5 Interpretation for practical use

MAE is understandable as average absolute error in USD, while RMSE adds sensitivity to occasional large misses. Both are needed here because the training distribution includes a relatively small number of costly properties and extreme observations. On the held-out set, the original forest shows useful predictive ability but notable errors at the top of the distribution. Its generalization is therefore assessed not only by R^2 but also by the absolute error pattern, with the optimization stage seeking to improve the balance between fitted and unseen performance.

6. Optimization Results

6.1 Three-fold cross-validation and grid-search setup

The original Random Forest was optimized using GridSearchCV with three-fold cross-validation on the 1,168-house training subset. Each candidate configuration was trained on two folds and evaluated on the remaining fold, with the process repeated across three folds. Negative Root Mean Squared Error (RMSE) was used as the scoring metric, and the best-performing configuration was refitted on the complete training subset.

The hyperparameter search examined eight configurations (2 × 2 × 2), resulting in 24 cross-validation fits. Hyperparameter selection was performed exclusively using the training data. The separate 292-house held-out test set was excluded from GridSearchCV and used to evaluate and compare the Linear Regression, original Random Forest, and optimized Random Forest models. This approach prevented direct leakage of held-out observations into hyperparameter fitting, although the reported performance remains specific to the selected data split.

Table 10. Search space used in the executed notebook

Hyperparameter	Values searched	Function
n_estimators	200, 300	Number of decision trees
min_samples_leaf	1, 2	Minimum examples allowed at terminal node
max_features	0.75, 1.0	Fraction of features considered at each split
max_depth	None (fixed)	No explicit maximum tree depth
min_samples_split	2 (fixed)	Minimum examples before splitting

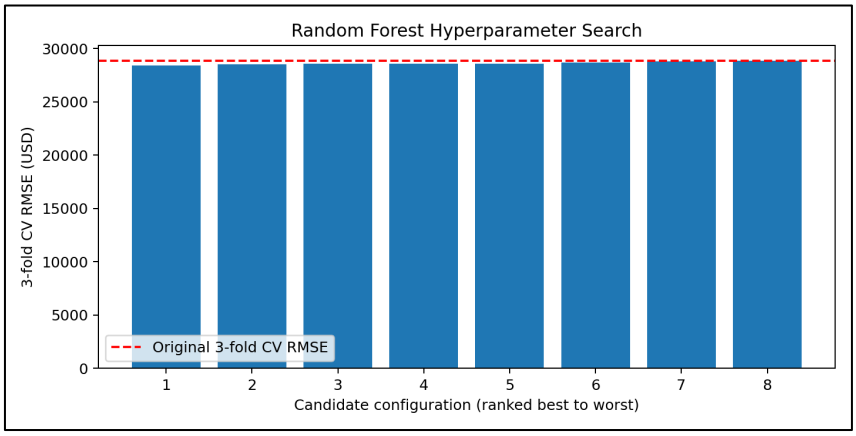


Figure 8. Ranked three-fold CV RMSE across all eight Random Forest candidates.

6.2 Cross-validation result and selection

The original forest had fold RMSEs of $29,796.22, $32,421.63 and $24,273.19 (mean $28,830.35; standard deviation $3,395.97). The best candidate achieved mean CV RMSE of $28,426.11, lower by $404.23 (approximately 1.40%). The winning settings were 300 trees, min_samples_leaf = 2 and max_features = 0.75, with max_depth = None and min_samples_split = 2 unchanged. The fold-to-fold variation is much larger than the mean improvement, so the result should be described as a limited gain rather than decisive evidence of large generalization improvement.

6.3 Held-out model comparison after tuning

Table 11. Baseline and optimized Random Forest on the same held-out houses

Metric	Original Random Forest	Optimized Random Forest	Change
MAE (USD)	17,433.97	17,224.73	−$209.24 (1.20%)
MSE (USD²)	865,449,435	853,619,317	−11,830,118 (1.37%)
RMSE (USD)	29,418.52	29,216.76	−$201.76 (0.69%)
R²	0.8872	0.8887	+0.0015

The optimized forest achieved MAE $17,224.73, RMSE $29,216.76 and R² 0.8887 on the 292-house held-out set. Test RMSE decreased by only 0.69% relative to the original forest. Figure 9 depicts that the difference is small in absolute bar height; Figure 10 still reveals several high-value houses whose predicted prices fall substantially below the perfect-prediction line. Thus, the search improved all four metrics, but it did not eliminate the largest errors or transform overall predictive accuracy.

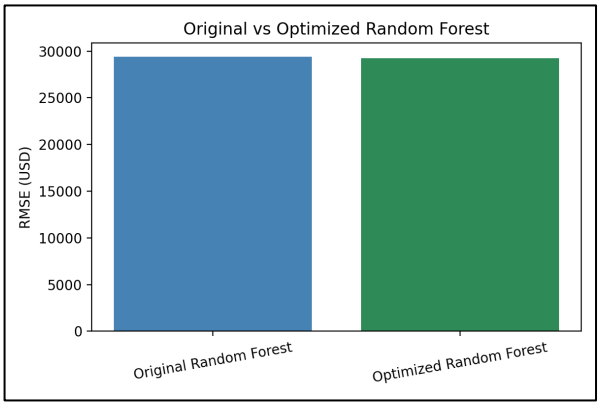


Figure 9. Held-out RMSE before and after tuning.

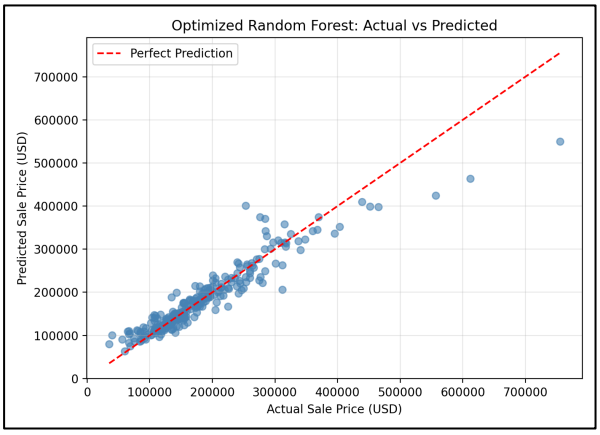


Figure 10. Optimized forest prediction agreement.

6.4 Generalization of the optimized model

Table 12. Optimized forest training versus held-out testing

Metric	Training	Held-out testing
MAE (USD)	6,942.82	17,224.73
RMSE (USD)	13,057.32	29,216.76
R²	0.9714	0.8887

The optimized model sacrificed some training fit compared with the original forest (baseline training MAE was $6,214.50) while slightly improving test results. That pattern is compatible with added regularization from the larger minimum leaf size and reduced max_features. However, the optimized training R² remains substantially higher than its testing R², so some generalization gap persists.

6.5 Feature importance and errors across price groups

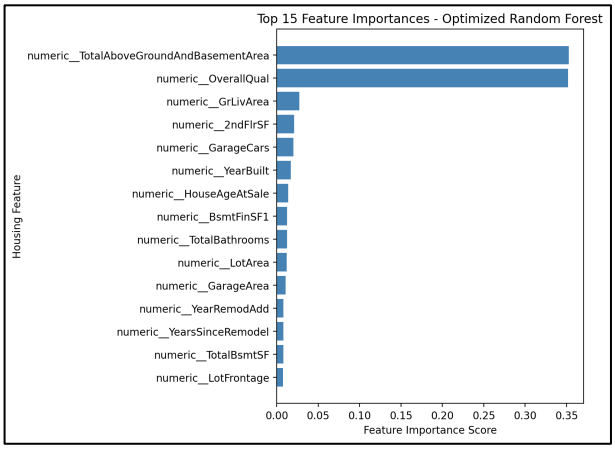


Figure 11. Leading impurity-based feature importances.

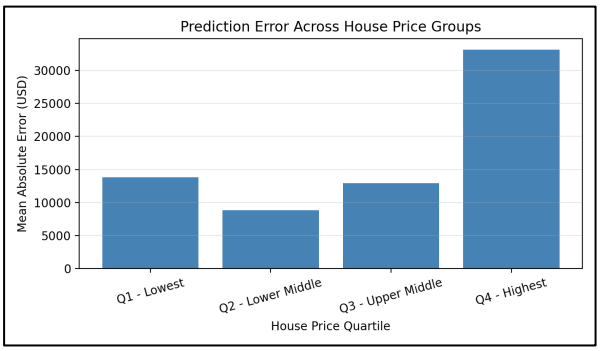


Figure 12. Mean absolute error by actual-price quartile.

Table 13. Optimized Random Forest: selected impurity-based feature importances

Feature	Importance	Interpretation
TotalAboveGroundAndBasementArea	0.3529	Engineered combined size measure
OverallQual	0.3518	Overall material/finish quality
GrLivArea	0.0274	Above-ground living area
2ndFlrSF	0.0212	Second-floor living area
GarageCars	0.0201	Vehicle capacity

TotalAboveGroundAndBasementArea and OverallQual account for approximately 70.5% of the model's total impurity-based feature importance. Their high importance is consistent with the relationships observed during EDA. However, some living-area features are strongly related to one another, which may affect how importance is distributed among them.

Table 14. Optimized test errors grouped by actual SalePrice quartile

Quartile	Houses	Mean actual price	MAE	Mean Absolute Percentage Error
Q1 lowest	75	$101,607	$13,824.83	17.66%
Q2 lower middle	71	$139,673	$8,854.94	6.31%
Q3 upper middle	73	$179,376	$12,926.64	7.24%
Q4 highest	73	$295,746	$33,156.34	10.37%

Figure 12 and Table 14 show that Q4 has the highest error in dollar terms ($33,156.34 MAE), whereas Q1 has the highest mean absolute percentage error (17.66%). The largest absolute errors therefore occur for high-price homes, but proportionally low-price homes also require attention. The quartiles contain slightly different sample counts because multiple houses share sale-price boundary values.

7. Conclusion and Future Improvements

7.1 Summary of outcomes

This assignment examined how machine learning could be used to predict house prices using the Kaggle Ames Housing dataset. The exploratory analysis showed that SalePrice was right-skewed, with OverallQual ($r = 0.791$) and GrLivArea ($r = 0.709$) having strong positive correlations with house prices. Several features contained missing values because certain facilities were absent from the properties. These values were handled according to their meaning, while numerical missing values were imputed and categorical variables were encoded. Additional features related to property age, living area, and facilities were also created. No duplicate records were found, and unusual properties were kept in the dataset after inspection.

Random Forest performed better than Linear Regression on the held-out test set. The original Random Forest achieved an RMSE of \$29,418.52 and an R^2 of 0.8872. After testing eight hyperparameter configurations using three-fold cross-validation, the best model used 300 trees, min_samples_leaf = 2, and max_features = 0.75. The optimized model achieved an RMSE of \$29,216.76, an MAE of \$17,224.73, and an R^2 of 0.8887. Although RMSE decreased by \$201.76 (0.69%), the improvement was relatively small.

7.2 Main limitations

- Training and testing performance: The optimized model achieved an R^2 of 0.9714 on the training set but only 0.8887 on the test set. This difference suggests that the model fitted the training data more closely than unseen observations.
- Differences across house-price groups: Prediction errors were not equally distributed. The highest-priced group (Q4) had an MAE of \$33,156.34, compared with \$8,854.94 for Q2. Meanwhile, Q1 had the highest mean absolute percentage error. These differences show why overall evaluation metrics alone may not fully describe model performance.
- Dataset limitations: The dataset only represents houses in Ames, Iowa, and includes relatively few expensive properties. The model was evaluated using one held-out test split and three-fold cross-validation for tuning. Its performance on properties from other locations or time periods has not been tested.
- Feature interpretation: Correlation coefficients and Random Forest feature-importance scores help identify relationships within the data, but they do not prove that a particular feature directly causes a change in house prices.

7.3 Future improvements

- Examine properties with unusually large living areas or high sale prices to understand why the model makes larger prediction errors for these houses.
- Explore additional hyperparameter combinations and use more cross-validation folds to examine whether the model performs consistently across different training subsets.
- Investigate feature selection and permutation importance to understand whether highly correlated features, particularly those related to property size, affect model performance and interpretation.
- Test the model using housing data from other locations or time periods to assess whether its predictions remain reliable beyond the Ames dataset.

7.4 Overall conclusion

The optimized Random Forest performed slightly better than the original model and produced reasonably accurate predictions for many houses in the held-out test set. However, prediction errors remained larger for

some expensive properties, and the difference between training and testing performance suggests that the model could still be improved. Further work should focus on reducing these errors and testing the model on different housing datasets before considering its use in other property markets.

References

[1] Kaggle, “House Prices: Advanced Regression Techniques,” dataset and competition materials. Available: https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/data (accessed November 2026).

[2] F. Pedregosa *et al*., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. [Online]. Available: https://www.jmlr.org/papers/v12/pedregosa11a.html.

[3] Scikit-learn Developers, “Pipelines and composite estimators,” *Scikit-learn Documentation*. [Online]. Available: https://scikit-learn.org/stable/modules/compose.html. [Accessed: Nov. 9, 2026].

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Appendix A. Source Code and GitHub Repository

The complete Python source code, original dataset, generated visualizations, and experimental results for this assignment are available in the GitHub repository below.

GitHub Repository: https://github.com/wenssy-gif/MAI5043-House-Price-Prediction

The repository provides supporting evidence of the implemented machine learning workflow and allows the source code and experimental outputs presented in this report to be reviewed.